



Biomolecules

- **G.N. Ramachandran**: triple helical structure of collagen
- **Linus Pauling**: explained α -helix and β -sheet structures
- The concepts and techniques of physics and chemistry are applied to understand biology. This Physico-chemical approach to study and understand living organisms is called '**Reductionist Biology**'.

Components of Living organism

How to find **organic** component of living organism?

Take living tissue $\rightarrow$ grind it in trichloroacetic acid $\rightarrow$ strain through cotton $\rightarrow$ you get filtrate (acid-soluble pool) and retentate or the (acid-insoluble fraction).

Acid soluble pool (Filtrate)	Acid insoluble pool (retentate)
Low Molecular weight (18 - 800 Dalton)	High Molecular weight (Molecular weight More than 10000 Dalton)
Biomacromolecules	Biomacromolecules
Sugar, amino acids, Nucleotides, ions	Carbohydrate, Proteins, Nucleic acid
Represent cytoplasmic composition	Represent cytoplasmic + cell organelle composition

- Weight of lipids do not exceed 800 Da, still considered as macromolecule because lipids form vesicles in acid insoluble pool
- Composition of cell :
Water (70%) > Protein (10%) > Carbohydrates (3%) > Lipids (2%)

How to find **inorganic** components of living organism?

Destructive method

Take living tissue (e.g. leaf) $\rightarrow$ dry it : water evaporates $\rightarrow$ burn it $\rightarrow$ carbon compounds gets oxidised to gaseous form $\rightarrow$ ash contains inorganic elements (like calcium) $\rightarrow$ do analysis

Carbohydrates

Cellulose

- Homopolymer i.e., polysaccharide consisting of only one type of monosaccharide e.g., Glucose
- Found in cell wall of plant, algae
- Cellulose is the most abundant organic polymer on Earth.
- Ruminants can digest cellulose
- Microbes in ruminants have Cellulase enzyme
- Humans can't digest cellulose
- Paper pulp uses cellulose

Starch

- store house of energy in plant tissues
- Starch forms helical secondary structures.
- Starch can hold I₂ molecules in the helical portion.
- The starch-I₂ is blue in colour.

Glycogen

- store house of energy in animal tissues

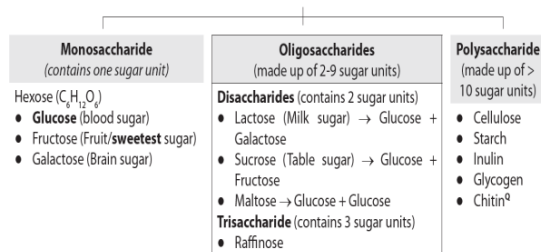
Inulin

- polymer of fructose

Chitin

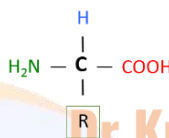
- Chitin is a complex polysaccharide
- Homopolymer of N acetyl Glucosamine
- Forms exoskeleton of insect and fungi

In a polysaccharide chain (say glycogen), the right end is called the reducing end and the left end is called the non-reducing end.



Amino acids

- Amino acids are called as α -amino acids.
- They are substituted methane
- Made up of hydrogen, carboxyl group, amino group and a variable group designated as R group.
- Around 20 types of amino acids are found in proteins



R group	Amino acid
Hydrogen	Glycine
Methyl group	Alanine
hydroxy methyl	Serine

- **Nature of amino acids**
 - ★ Acidic amino acids $\rightarrow$ Glutamic acid
 - ★ **Basic** amino acid $\rightarrow$ Lysine *Uska basic loose hai*
 - ★ Neutral amino acid $\rightarrow$ Valine
 - ★ aromatic amino acids $\rightarrow$ tyrosine, phenylalanine, tryptophan
- **Zwitter ions** *PT karneka try karo aroma niklega*
 - ★ A particular property of amino acids is the ionizable nature of $-\text{NH}_2$ and $-\text{COOH}$ groups. Hence in solutions of different pH, the structure of amino acids changes. Under neutral conditions, the amino acid will exist in its zwitterion form. A zwitterion is a molecule that contains both a positive and a negative charge, pH is called isoelectric pH
- **Essential amino acids**
 - ★ amino acids which are essentially supplied through our diet are called essential amino acids
- **Non-essential amino acids**
 - ★ These are amino acids that body can make

Proteins

- Proteins are polypeptides.
- They are linear chains of amino acids linked by peptide bonds
- Protein is a heteropolymer and not a homopolymer
- **Collagen** is the most abundant protein in animal world
- Ribulose biphosphate Carboxylase-Oxygenase (**RuBisCO**) is the most abundant protein in the whole of the biosphere.

Protein	Functions
Collagen	Interacellular ground substance
Trypsin	Enzyme
Antibody	Fights infectious agents
Receptor	Sensory reception (smell, taste, hormone, etc.)
GLUT-4	Enables glucose transport into cells

Structure of proteins

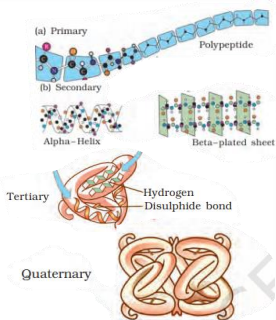
Isko mat bhulna

Primary structure	Amino acids linked with each other to form thread like structure
Secondary structure	The amino acid thread, fold upon itself to form alpha helix and beta pleated sheet . In proteins, only right-handed helices are observed
Tertiary structure	Three-dimensional folding pattern where protein thread is folded upon itself (like a hollow woollen ball) with the help of hydrogen and disulphide bond. Tertiary structure is absolutely necessary for the many biological activities of proteins
Quaternary structure	Protein consisting of more than one amino acid chain e.g., Haemoglobin



TABLE 9.3 Some Secondary Metabolites

Pigments	Carotenoids, Anthocyanins, etc.
Alkaloids	Morphine, Codeine, etc.
Terpenoids	Monoterpenes, Diterpenes etc.
Essential oils	Lemon grass oil, etc.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, etc.
Polymeric substances	Rubber, gums, cellulose

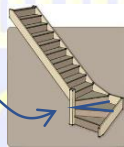


Primary and secondary metabolites

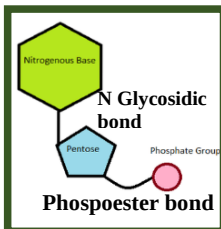
- Primary metabolites
 - Primary metabolites are molecules which have identifiable functions and known roles in normal physiological processes,
 - e.g., amino acids, sugars, etc.
- Secondary metabolites
 - Secondary metabolites are molecules produced by plant, fungal and microbial cells whose role or functions are not known.

Nucleic acid

- DNA is a macromolecule
- Nucleic acids exhibit secondary structures
- DNA double helix model was proposed by Watson and Crick
- DNA is a polymer of nucleotide
- Nucleotides are made up of 3 components: monosaccharide, Phosphoric acid or phosphate and heterocyclic compound
- Heterocyclic ring is Purine (Adenine and Guanine) and Pyrimidines (Cytosine, thymine, uracil)
- Each strand appears like a helical staircase.
- At each step of ascent, the strand turns 36°.
- One full turn of the helical strand - ten base pairs.
- The pitch = 34Å.
- The rise per base pair = 3.4Å. This
- B-DNA is the term given for the **right-handed DNA helix that is the most common form of DNA**



DNA bonds



Pentose sugar + Nitrogen base: Nucleoside
Nucleoside + Phosphate group: Nucleotide
5' end = free phosphate group
3' end = free OH group

RNA

Sugar: Ribose sugar (- OH group at 2' C)
Uracil in place of thiamine (5 Methyl Uracil)

Nucleoside	Nucleotides
Adenosine	Adenylic acid
Guanosine	Thymidylic acid
Thymidine	Guanidylic acid
Uridine	Uridylic acid
Cytidine	Cytidylic acid

Lipids

- Lipids are: fatty acids, glycerol and phospholipids

Fatty acids

- A fatty acid has a carboxyl group attached to an R group
- Palmitic acid has 16 carbons, Arachidonic acid has 20 carbon
- Types
 - Saturated Fatty acids: no double bond
 - Unsaturated fatty acids: have one or more C=C double bonds

Glycerol

- Glycerol is trihydroxy propane
- Many lipids have both glycerol and fatty acids. Here the fatty acids are found esterified with glycerol

Phospholipid

- Phospholipids have phosphorous in them
- E.g., Lecithin: major component of bilayer cell membrane
- Sphingolipid: found in myelin sheath of nervous cells

Bonds in Polymers

Polymer	Monomer	Bonds in monomer
Polysaccharide	Monosaccharide	Glycosidic bonds
Proteins	Amino acids	Peptide Bonds
DNA	Nucleotides	Phosphodiester bonds

Lipid	Lipid is not a polymer
-------	------------------------

Dynamic state of body constituents

- The metabolic pathway includes a series of reactions. The metabolite flow, the rate, and direction at which metabolism takes place are called the **dynamic state of body constituents**.
- All metabolic reactions are catalysed by a set of proteinaceous compounds called **enzymes**.
- Anabolic pathways:
 - Simple structure form complex structure
 - Consumes energy
 - E.g., acetic acid becomes cholesterol
- Catabolic pathway:
 - Complex structure leads to simpler structure
 - Releases energy
 - E.g., Glucose becomes lactic acid in our skeletal muscle

Anabolism means constriction & catabolism means destruction

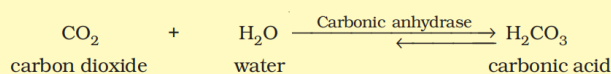
The living state

- The living state is a **non-equilibrium** steady-state to be able to perform work.
- The living process is a constant effort to prevent falling into equilibrium and is achieved by energy input. The metabolism provides a mechanism for the production of energy.
- The living state and metabolism are synonymous, without metabolism there cannot exist a living state.
- A living organism works continuously so it cannot afford to reach equilibrium because a system at equilibrium cannot perform the work



Enzymes

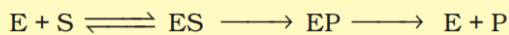
- Almost all enzymes are proteins
- Except ribozyme which is a nucleic acid (RNA)
- The **Globular tertiary structure** enables proteins to function as enzymes.
- An active site of enzyme is a pocket into which the substrate fits
- Difference between Enzyme catalysts and inorganic catalysts
 - Inorganic catalysts work efficiently at high temperatures and high pressures, while enzymes get damaged at high temperatures
- Carbonic anhydrase enzyme (found in RBC and Plasma) accelerates the below reaction rate by about 10 million times





Mechanism of enzyme action

- Enzyme bring out chemical and physical changes in substrate
- Enzyme substrate complex is a transient state
- In both exothermic and endothermic reaction, the 'S' has to go through a much higher energy state or transition state means substrate needs to be activated by energy called as activation energy
- Activation energy is needed in both Endothermic and exothermic reaction
- Role of enzyme in reactions is to **decrease** activation energy



The catalytic cycle of an enzyme action can be described in the following steps:

1. First, the **substrate binds** to the active site of the enzyme,
2. Enzyme alter its shape and **fit more tightly around the substrate**.
3. Enzyme **breaks** the chemical bonds of the substrate
4. The enzyme **releases** the products of the reaction and
5. The free enzyme is **ready to bind to another molecule** of the substrate

Factors Affecting Enzyme Activity

- (1) Temperature and pH
- (2) substrate concentration
- (3) binding of specific chemicals

(1) Temperature and Ph

- Each enzyme shows its highest activity at a particular temperature and pH called the optimum temperature and optimum pH
- Activity declines both below and above the optimum value.
- Low temperature → preserves enzyme in a temporarily inactive state
- high temperature → destroys enzymatic activity because proteins are denatured by heat.

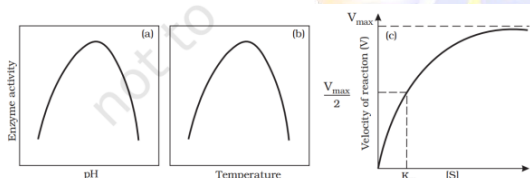


Figure 9.7 Effect of change in : (a) pH (b) Temperature and (c) Concentration of substrate on enzyme activity

- ✓ Km is Substrate conc at which enzyme velocity is half of Vmax
- ✓ Vmax is the maximum velocity of the reaction for the given concentration of enzyme.

(2) Concentration of Substrate

With the increase in substrate concentration, the velocity of the enzymatic reaction rises at first. The reaction ultimately reaches a maximum velocity (Vmax) which is not exceeded by any further rise in concentration of the substrate

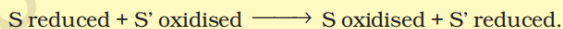
(3) binding of specific chemicals

- **Inhibitor non competitive bhi ho sakta hai par NCERT ne competitive inhibitor hi apko bataya hai**
- closely **resembles the substrate**
- competes with substrate for substrate binding site of the enzyme
- Consequently, the substrate cannot bind and as a result, the enzyme action declines
- e.g., inhibition of **succinic dehydrogenase by malonate** which closely resembles the substrate succinate in structure.
- Such competitive inhibitors are often used in the control of bacterial pathogens e.g., Penicillin, for example, is a competitive inhibitor that blocks the active site of an enzyme that many bacteria use to construct their cell wall

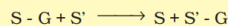
Classification and Nomenclature of Enzymes

Enzymes are divided into 6 classes each with 4-13 subclasses and named accordingly by a four-digit number. **OT HILL**

Oxidoreductases/dehydrogenases: Enzymes which catalyse oxidation-reduction between two substrates S and S' e.g.,



Transferases: Enzymes catalysing a transfer of a group, G (other than hydrogen) between a pair of substrate S and S' e.g.,

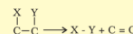


Hydrolases: Enzymes catalysing hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide or P-N bonds.

Isomerases: Includes all enzymes catalysing inter-conversion of optical, geometric or positional isomers.

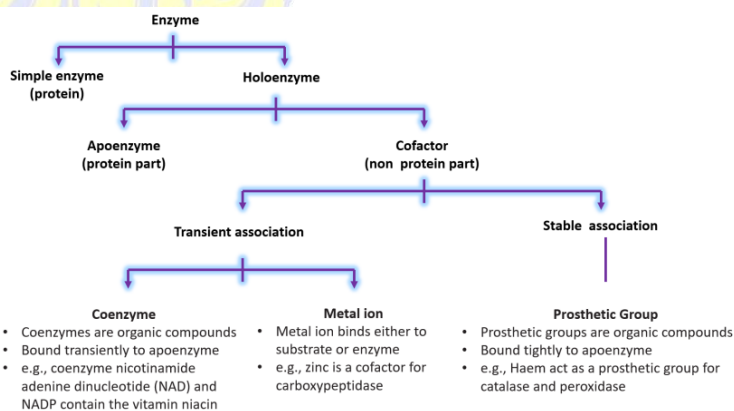
Ligases: Enzymes catalysing the linking together of 2 compounds, e.g., enzymes which catalyse joining of C-O, C-S, C-N, P-O etc. bonds.

Lyases: Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds



Co-factors

- Enzymes are composed of one or several polypeptide chains
- However, there are a number of cases in which non-protein constituents called cofactors are bound to the enzyme to make the enzyme catalytically active
- Catalytic activity is lost when the co-factor is removed from the enzyme
- The protein portion of the enzymes is called the apoenzyme
- Three kinds of cofactors may be identified: prosthetic groups, co-enzymes and metal ions.



-----END-----

<https://youtu.be/WPjg6osSZL8?t=80>

